Center-Guided Classifier for Semantic Segmentation of Remote Sensing Images

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***Abstract*—Compared with natural images, remote sensing images (RSIs) have a unique characteristic, i.e., larger intraclass variance, which makes semantic segmentation for RSIs more challenging. Moreover, existing semantic segmentation models for RSIs usually employ a vanilla softmax classifier, which has three drawbacks: 1) nondirect supervision for the pixel representations during training; 2) inadequate modeling ability of parametric softmax classifiers under large intraclass variance; and 3) an opaque process of classification decision. In this article, we propose a novel classifier (called CenterSeg) customized for RSI semantic segmentation, which solves the above-mentioned problems with *multiple prototypes*, *direct supervision under Grassmann manifold*, and *interpretability strategy*. Specifically, for each class, our CenterSeg obtains local class centers by aggregating corresponding pixel features based on ground-truth masks and generates multiple prototypes through hard attention assignment (HAA) and momentum updating. In addition, we introduce the Grassmann manifold and constrain the joint embedding space of pixel features and prototypes based on two additional regularization terms. Especially during the inference, CenterSeg can further provide interpretability to the model by restricting the prototype to a sample of the training set. Experimental results on three remote sensing segmentation datasets validate the effectiveness of the model. Besides the superior performance, CenterSeg has the advantages of simplicity, lightweight, compatibility, and interpretability. Code is available at https://github.com/xwmaxwma/rssegmentation**

***Index Terms*—Class center, classifier, prototype, remote sensing images (RSIs), semantic segmentation.**

I. Introduction

SEMANTIC segmentation of remote sensing images (RSIs) is a fundamental task in the remote sensing community for parsing and understanding RSIs obtained from satellite or aerial photography [1], which benefits various real-world applications, including environmental monitoring [2], urban planning [3], and agricultural management [4]. The key objective of RSI semantic segmentation is to segment the RSI into different regions/objects and assign each of them with a specific category (e.g., water, forests, and buildings) [5], [6].

In the early days, some RSI segmentation methods based on specific scenes (e.g., road/building segmentation [7], [8]) follow the traditional segmentation framework [9], [10], [11], [12], [13]. They mainly focus on improving special application scenarios but frequently ignore the characteristics of RSI themselves, which limits their application scopes. Recently, generic RSI semantic segmentation methods have been widely investigated, which aim to segment multiple land cover classes [14], [15], [16].

Compared to images captured in natural scenes [17], the unique characteristic of RSI makes accurate segmentation difficult, i.e., different scenes and different areas of the same scene have significant intraclass variance due to complex spatial distributions. As illustrated in Fig. 1, the red boxes in Scene 1, the yellow boxes in Scene 2, and the red and yellow boxes between the two scenes exhibit diverse colors and morphologies among themselves, yet share the same semantic (i.e., building), underscoring large intraclass variance.

A series of effective approaches have been successively proposed to mitigate the intraclass variance by optimizing the backbone's design [14] and context modeling [18], [19], [20], [21]. However, these RSI semantic segmentation models commonly use parametric softmax classifiers, making these approaches frequently face the following limitations: 1) parametric softmax classifiers use only one weight vector with bias to represent a specific class, which usually has large intraclass variance, which limits the performance of parametric softmax classifiers; 2) the parametric softmax classifier's cross-entropy loss function optimizes the parameters by supervising the classification probability of each pixel instead of directly supervising the pixel representation, which prevents the pixel representation from being well-trained; and 3) the weight matrix and the bias vector are both learnable parameters, where the decision-making process of their classifications is not transparent, which makes it difficult to provide interpretability for these models.

Received 9 May 2025; revised 15 September 2025 and 14 December 2025; accepted 12 January 2026. Date of publication 28 January 2026; date of current version 12 March 2026. This work was supported in part by the Public Welfare Science and Technology Plan of Ningbo City under Grant 2022S125 and in part by the Key Research and Development Plan of Zhejiang Province under Grant 2021C01031. *(Corresponding author: Xiaowen Ma.)*

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Digital Object Identifier 10.1109/TGRS.2026.3657453

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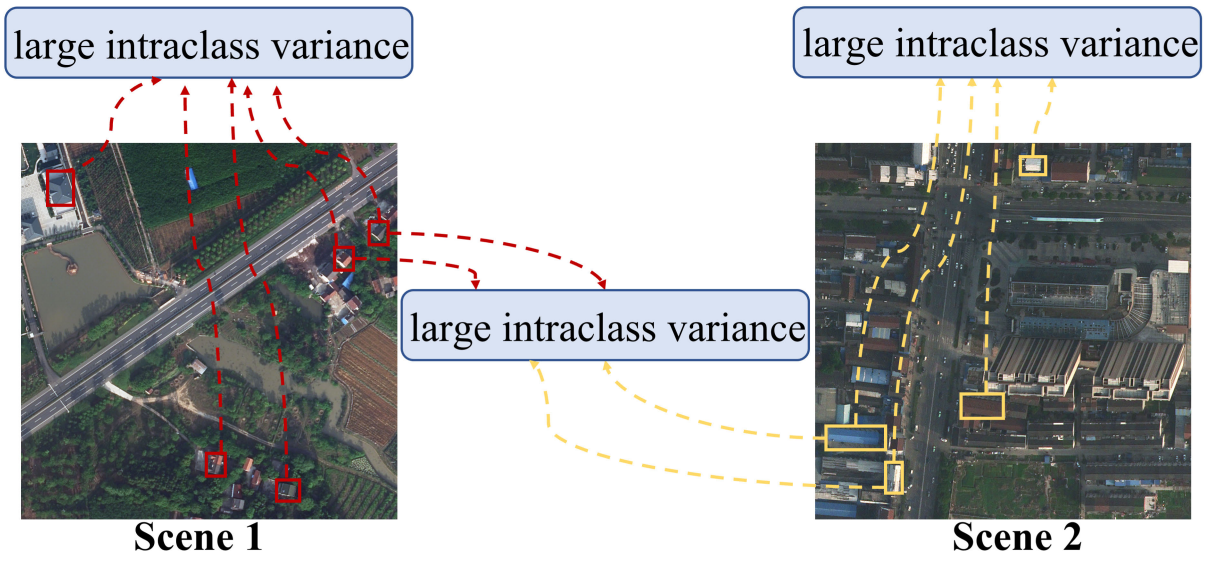


Fig. 1. Display of the RSI characteristic. Images are selected from the LoveDA dataset. For example, the color and shape of the building class are significantly different between Scenes 1 and 2, as well as between regions of the same scene, underscoring the characteristic of RSIs, i.e., large intraclass variance.

Some strategies [22], [23] have been used to improve softmax classifiers on semantic segmentation for natural images, however, directly applying these prototyping methods (e.g., ProtoSeg [24]) to high-resolution RSIs faces significant challenges: their reliance on external storage and online clustering mechanisms incurs high computational and memory overhead, and their prototype generation process fails to adequately adapt to the complex intraclass structure and multiscale characteristics of remotely sensed features. To address these challenges, this article proposes a novel prototype learning and inference paradigm specifically designed for RSIs–CenterSeg. Unlike existing paradigms that require runtime prototype computation, the core idea of CenterSeg is to learn a set of static, parameterized class centers. Technically, the local clustering strategy is innovatively introduced to generate representative feature samples, i.e., class centers, which can be used to accelerate model training and reduce the memory. Specifically, CenterSeg applies the ground-truth to guide feature aggregation during training, enforcing the network to generate local class centers, which are used to generate multiple semantically accurate prototypes based on hard attention assignments (HAAs) and momentum updates. The prototypes will remain fixed after the training phase concludes and will be stored within the model parameters. At the inference stage, our CenterSeg directly applies these prototypes to perform classification decisions. Furthermore, recent studies [25] have demonstrated that intraclass compactness and interclass separability can enhance the performance of semantic segmentation. Therefore, to optimize the generation of prototypes, we introduce the Grassmann manifold and propose two additional regular terms to constrain the joint embedding space of pixel features and prototypes. To the best of our knowledge, our CenterSeg is the first approach to design a new classifier to improve the RSI semantic segmentation task with the following advantages.

1) *Effectiveness:* Due to the introduction of multiple prototypes and regularization constraints, CenterSeg is able to cope well with the large within-class variance challenge and has a good embedding space of pixel features with prototypes. We conducted extensive experiments on three RSI semantic segmentation datasets. The results show that CenterSeg can significantly improve the segmentation performance of various baseline models.
2) *Simple:* CenterSeg does not require a lot of extra storage and computation consumption during training, which makes it easy to be applied and practically deployed.
3) *Lightweight:* CenterSeg is tested based on the prototypes obtained during the training process, without introducing additional parameters, and the increase in computational resources is negligible.
4) *Compatible:* CenterSeg is compatible with existing RSI semantic segmentation methods, and only needs to replace the vanilla softmax classifier.
5) *Interpretable:* If the prototype is restricted to samples (images) from the training set, CenterSeg can apply *this looks like that* rule [26] to interpret its predictions, which promotes the interpretability of the model.

II. Related Works

A. Semantic Segmentation

Semantic segmentation is a fundamental vision task, which aims to achieve pixel-level classification of input images [27]. Recently, several works have focused on the segmentation of RSI [28]. Different from natural images, RSIs have several unique data characteristics [29], i.e., complex backgrounds and large intraclass variance. To address the problem, LANet [30], SCSM [31], PointFlow [32], SCO [33], Farseg++ [16], and LOGCAN++ [19] design the customized network to seek high-quality segmentation results. However, these methods all rely on a softmax classifier, which performs poorly on a dataset with large intraclass variance. In this article, we start from the classifier perspective and design a plug-and-play classifier. As we all know, this is the first customized classifier for RSI segmentation.

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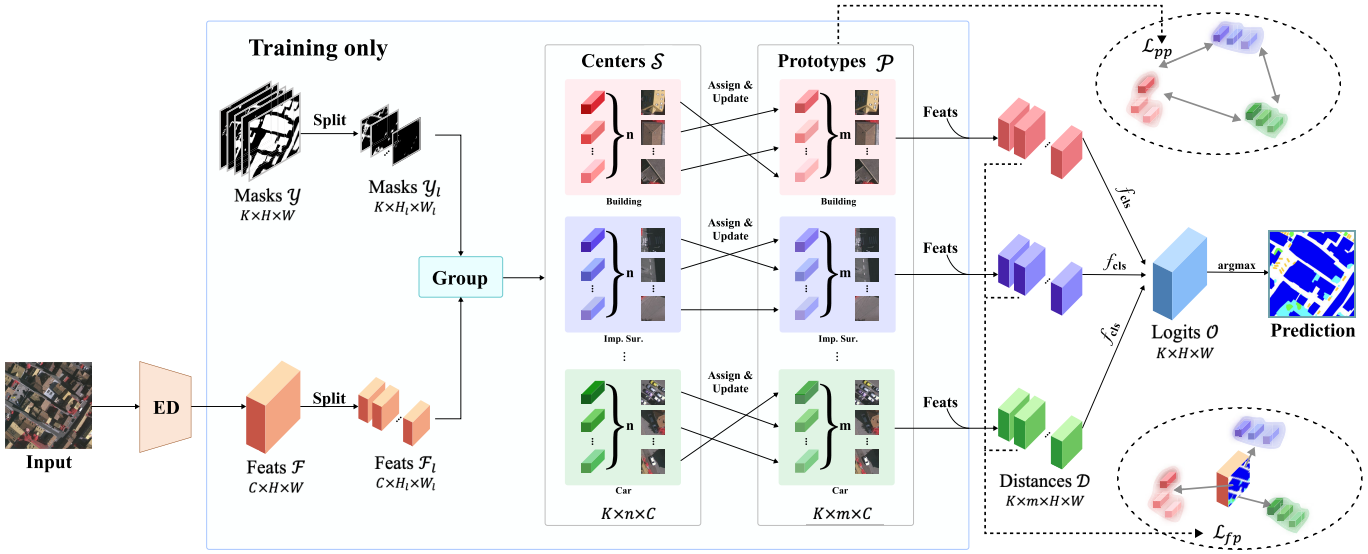


Fig. 2. Architecture of the proposed CenterSeg, which consists of two key components: prototype generation and regularizer terms. At the training stage, the input image is first projected to a high-dimensional space after the ED to obtain the pixel features $\mathcal{F}$. Next, we aggregate the features based on the ground-truth mask to obtain the class centers $\mathcal{S}$, i.e., the representative features of each class. Then, we generate the prototype $\mathcal{P}$ based on HAA and momentum update. In addition, two regularizer terms $\mathcal{L}_{pp}$ and $\mathcal{L}_{fp}$ are proposed to optimize the prototype generation. At the inference stage, classification decisions are performed directly based on the similarity of the pixel features to the prototype.

B. Classifier

In response to the limitations of parametric softmax classifiers, several dynamic prototype-based paradigms have been proposed for semantic segmentation in natural images [22], [23], [24], [25]. These methods share a common paradigm: they rely on an external memory mechanism and online clustering or retrieval processes during inference to generate or match class prototypes. For instance, ProtoSeg [24] pioneers a nonparametric scheme using unlearnable prototypes derived from average features, requiring nearest-neighbor searches for prediction. GMMSeg [23] models class distributions with Gaussian mixture models, while DNC [22] interprets classification as proximity to class submasses. SSA-Seg [25] adapts prototypes to the test image's domains. Although effective, this dynamic paradigm faces inherent challenges when applied to high-resolution RSIs: 1) the external storage and continuous online clustering incur substantial computational and memory overhead, conflicting with the efficiency requirements of processing large-scale RSIs; and 2) the prototype generation, often based on global feature statistics, may not adequately capture the extreme intraclass variance and complex multimodal structures characteristic of RSIs (e.g., diverse building forms and varying ship scales).

Departing from this dynamic paradigm, we propose a novel static prototype learning framework specifically engineered for RSI segmentation. Our core insight is to internalize the prototype representation as static, learnable parameters of the model itself, eliminating the need for any external memory or runtime clustering. CenterSeg achieves this through two pivotal designs. First, it employs ground-truth-guided local feature aggregation during training to generate semantically precise class centers, which are then solidified via momentum updates. This ensures prototypes are representative of the local contextual patterns prevalent in RSIs. Second, and more distinctively, we introduce a Grassmann manifold-based regularization to explicitly shape the joint embedding space of pixel features and prototypes, enforcing intraclass compactness and interclass separability—a systematic optimization not addressed in prior prototype methods. Consequently, CenterSeg transitions the prototype from a dynamically computed artifact to a statically stored parameter, resulting in a model that is more efficient to deploy, more directly aligned with RSI characteristics, and provides a new, interpretable pathway for feature-protype space optimization.

III. Methodology

This article proposes CenterSeg, an end-to-end static prototype learning framework tailored for RSIs. As shown in Fig. 2, the overall architecture includes three core innovative designs: 1) a truth-guided local prototype generator that dynamically aggregates pixel features to form representative class centers; 2) a manifold-constrained prototype optimizer that systematically improves the geometric properties of the feature-prototype space through Grassmann regularization; and 3) a parameterized prototype classification head where optimized prototypes are permanently stored for efficient inference. Unlike dynamic prototyping methods, the prototypes learned by this framework are part of the final model and do not require external system support.

A. Motivation

Typical RSI semantic segmentation models [16], [19] usually consist of three parts: an encoder, a decoder, and a classifier. Given an input image, the encoder and decoder (ED) project it to a high-dimensional feature $\mathcal{F} \in \mathbb{R}^{C\times H\times W}$, where C, H, and W denote the channel, height, and width of it, respectively. The classifier then acts as a feature descriptor whose weights are used as decision boundaries in the high-level feature space, describing the feature distribution, and

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performing the category judgment. Previous approaches [14], [16], [18] built on a parameterized softmax classifier apply a 1×1 convolution to obtain a vector of probability distributions $\mathcal{O}_i$ for each pixel i, and then produce the classification result $\hat{\mathcal{O}}$ by selecting the category with the highest probability. This process is formulated as

$$\mathcal{O}_i = \mathbf{W}^T \mathcal{F}_i + \mathbf{b}, \hat{\mathcal{O}}_i = \arg\max_k (\mathcal{O}_i) \tag{1}$$

where $\mathbf{W} \in \mathbb{R}^{K \times C}$ and $\mathbf{b} \in \mathbb{R}^K$are the learnable weight matrix and bias, respectively, and $\mathcal{F}_i \in \mathbb{R}^C$ is the feature vector of pixel i. $\arg\max_k$ denotes the argmax operation in the category dimension. The weights of the model are then optimized based on the cross-entropy loss as

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^{N} \sum_{k=1}^{K} y_i^k log\left(\mathcal{O}_i^k\right) \tag{2}$$

where N denotes the total number of pixels, K is the number of categories, and y_i^k denotes the ground-truth label of the pixel i. As demonstrated in (1) and (2), the softmax classifier represents each class only based on a single learnable weight and bias (i.e., a prototype) that cannot tolerate high intraclass variance. Furthermore, it optimizes the parameters by supervising the relative classification probability of each pixel, rather than directly supervising the pixels' semantic representations (which cannot effectively constrain the embedding space of features). This motivates us to propose our CenterSeg, a center-guided classifier, to address the above issues.

B. CenterSeg Overview

As shown in Fig. 2, our CenterSeg aggregates pixel features based on ground-truth masks to generate class centers $\mathcal{S}$ that characterize the semantic distribution of each class. Then, based on the obtained class centers $\mathcal{S}$, we generate class prototypes $\mathcal{P}$ by using HAA and momentum updating. Finally, the similarity between the pixel features $\mathcal{F}$ and each class prototype is computed, and a *winner-take-all* rule is applied, i.e., the segmentation result is determined by the class to which the most similar prototype belongs. This computational procedure can be formulated as

$$\hat{\mathcal{O}} = \arg\max_{m \times K} \left(f_{\text{cls}}\left(\|\mathcal{F} - \mathcal{P}\|_2\right)\right) \tag{3}$$

where $\mathcal{P}$ is the generated class prototypes, $m \times K$ represents the total number of class prototypes, $\|\mathcal{F} - \mathcal{P}\|_2$ denotes computing the Euclidean distance from the feature $\mathcal{F}$ to the prototype $\mathcal{P}$, and f_{cls} represents the similarity calculation function as

$$f_{cls}(t) = \frac{1}{1 + \alpha t}. \tag{4}$$

In addition, to optimize the generation of class prototypes, we introduce the Grassmann manifold and constrain the joint embedding space representing pixel features and class prototypes based on two additional regular terms (i.e., $\mathcal{L}_{pp}$ and $\mathcal{L}_{fp}$). Therefore, the loss function of CenterSeg is

$$\mathcal{L} = \mathcal{L}_{ce} + \lambda_{pp}\mathcal{L}_{pp} + \lambda_{fp}\mathcal{L}_{fp} + \lambda_{Dice}\mathcal{L}_{Dice} \tag{5}$$

where λ_{pp}, λ_{fp}, and λ_{Dice} are adjustable parameters. This way, multiple high-quality prototypes of each class can be obtained and further constrain the feature generation space to enhance the performance of existing RSI segmentation models [14], [15]. In Section III, we describe the generation process of the prototype (see Section III-C) and the regular term (see Section III-D) in detail.

C. Class Prototypes Generation

To generate semantically accurate and more discriminative prototypes, this step starts with directly aggregating pixel features based on ground-truth masks to generate class centers. Specifically, as shown in Fig. 2, it converts labels to one-hot format and then downsamples them to generate the ground-truth masks $\mathcal{Y} \in \mathbb{R}^{K \times H \times W}$ (K is the number of classes). To generate local class centers, the obtained $\mathcal{F}$ and $\mathcal{Y}$ are then split along the spatial dimension as

$$\mathcal{F}_l \in \mathbb{R}^{n \times C \times (H_l \times W_l)} = \text{Split}\left(\mathcal{F} \in \mathbb{R}^{C \times H \times W}\right) \tag{6}$$

$$\mathcal{Y}_l \in \mathbb{R}^{n \times K \times (H_l \times W_l)} = \text{Split}\left(\mathcal{Y} \in \mathbb{R}^{K \times H \times W}\right) \tag{7}$$

where H_l and W_l denote the height and width of the split patches, respectively. Consequently, the class centers $\mathcal{S} \in \mathbb{R}^{K \times n \times C}$ is obtained as

$$\mathcal{S} = \mathcal{Y}_l \otimes \mathcal{R}_l \tag{8}$$

where $n = (H/H_l) \times (W/W_l)$ is the number of patches.

We then assign the obtained class centers ($\mathcal{S}$) to their corresponding class prototype via the hard attention whose similarity matrix $\mathcal{A}$ is computed via a Gumbel-softmax [34] operation as

$$\mathcal{A}_k^{i,j} = \frac{\exp\left(\mathcal{S}_k^i \mathcal{P}_k^j + \gamma_i\right)}{\sum_{t=1}^{m} \exp\left(\mathcal{S}_k^t\right) \mathcal{P}_k^j + \gamma_t} \tag{9}$$

where $\mathcal{S}_k^i$ and $\mathcal{P}_k^j$ denote the ith center and jth prototype of the class k, respectively, and γ_i are i.i.d. random samples drawn from the Gumbel (0, 1) distribution. Since the hard-attention one-hot assignment operation via argmax is not differentiable, we apply the straight-through trick in [35] to compute the assignment matrix as

$$\hat{\mathcal{A}} = \text{one-hot}\left(\mathcal{A}_{\text{argmax}}\right) + \mathcal{A} - \psi(\mathcal{A}) \tag{10}$$

where ψ denotes the stop gradient operation. As shown in (10), $\hat{\mathcal{A}}$ has the one-hot value of assignment to a single group, whose gradient is equal to the gradient of $\mathcal{A}$. We then perform the assignment as

$$\hat{\mathcal{P}}_k^i = \frac{1}{\mathcal{N}(i)} \sum_j \hat{\mathcal{A}}_k^{i,j} \otimes \mathcal{S}_k^j. \tag{11}$$

Therefore, the prototypes of the current minibatch $\hat{\mathcal{P}}$ can be obtained, which are then used to update the final prototypes $\mathcal{P}$ in momentum

$$\mathcal{P} = \mu\mathcal{P} + (1 - \mu)\hat{\mathcal{P}} \tag{12}$$

where $\mu \in [0, 1]$ is the momentum update coefficient.

D. Regularizer Term

To optimize the generation of prototypes, we introduce the Grassmann manifold based on two additional regular terms (i.e., $\mathcal{L}_{pp}$ and $\mathcal{L}_{fp}$), targeting at constraining the joint embedding space of pixel features and prototypes.

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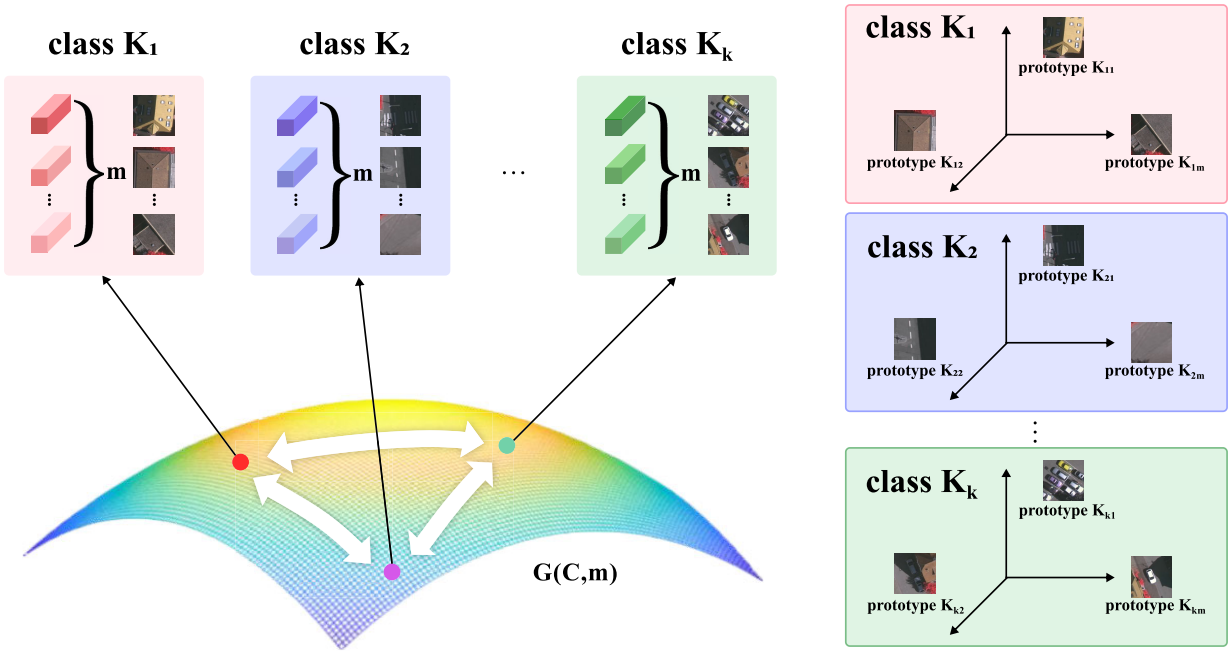


Fig. 3. Illustration of Grassmann manifold semantic space. (Left) Space is constructed by category-aware basis concepts, where the subspace of each class could be regarded as a point on the Grassmann manifold. (Right) Basis of subspaces representing classes are orthogonal to each other. By minimizing the distance between the projection matrices of two points on the Grassmann manifold, subspaces of different classes can be made far away from each other, thereby achieving better class separation.

1) Prototype-to-Prototype Regularizer: Our goal is to obtain diverse class prototypes that favor tolerance of large intraclass variance. This means each class is represented by a set of prototypes, each of which should have its own unique representation. Thus, all subprototypes of each class can be regarded as a set of basis vectors that should not be overlapped, i.e., they need to be orthogonal. To achieve this, we introduce an orthogonal idea to design the prototype-to-prototype loss $\mathcal{L}_{pp}$ based on the Grassmann manifold semantic space (i.e., this space has good orthogonal properties).

a) Intraclass prototype orthogonality: The prototype-to-prototype regularizer introduces orthogonal constraints to each prototype intraclass, thereby preventing the class prototype $\mathcal{P}$ degeneration from $m \times K$ class prototypes to K class prototypes

$$\mathcal{L}_{pp_1} = \frac{1}{K}\sum_{k=1}^{K}\left\|\mathcal{P}_k^T\mathcal{P}_k - I_m\right\|_F^2 \tag{13}$$

where $\mathcal{P}_k \in \mathbb{R}^{C\times m}$ is the subprototypes of k_{th} class, $|\cdot\|_F$ represents *Frobenius* norm, and I_m is the identity matrix.

b) Orthogonalized the subspaces of different classes: As shown in Fig. 3, each point on the Grassmann manifold corresponds to a unique projection matrix. By using a proper projection function, vectors in different subspaces can be projected onto the corresponding subspace to obtain representation in the subspace [36]. Inspired by this, the subspace of each class could be regarded as a point on the Grassmann manifold in CeterSeg. In Grassmann manifolds, a common subspace distance metric is the projection metric, which can be expressed as

$$\phi(\mathcal{P}_{k1},\mathcal{P}_{k2}) = \frac{1}{\sqrt{2}}\left\|E(\mathcal{P}_{k1})E(\mathcal{P}_{k1})^T - E(\mathcal{P}_{k2})E(\mathcal{P}_{k2})^T\right\|_F. \tag{14}$$

By minimizing the distance between the projection matrices of two points on the Grassmann manifold, subspaces of different classes can be made far away from each other, thereby achieving better class separation. Thus, the loss function can be defined by maximizing the metric of the projection between each pair of subspaces

$$\mathcal{L}_{pp_2} = \sum_{k1=1}^{K-1}\sum_{k2=k1+1}^{K}\phi(\mathcal{P}_{k1},\mathcal{P}_{k2}). \tag{15}$$

As a result, the final prototype-to-prototype loss function ($\mathcal{L}_{pp}$) can be expressed as

$$\mathcal{L}_{pp} = \mathcal{L}_{pp_1} + \mathcal{L}_{pp_2}. \tag{16}$$

Therefore, the diversity of intraclass prototypes and the distance of interclass prototypes are obtained by $\mathcal{L}_{\text{pp}}$, which facilitates the construction of a solid prototype representation space.

2) Feats-to-Prototype Regularizer: The prototype-to-prototype regularizer simply constrains class prototypes $\mathcal{P}$. In this section, we design the feats-to-prototype regularizer to constrain extracted features $\mathcal{F}$.

A good joint embedding space of features and prototypes should satisfy two conditions: 1) the distance between the pixel features and the class prototype of the ground-truth class to which it belongs should be as small as possible and 2) the distance between the pixel features and the class prototype of the nonground-truth class to which it belongs should be as large as possible. Therefore, we design according to the above two objectives in order to directly supervise the pixel representation (the parameterized softmax classifier supervises the relative category probabilities).

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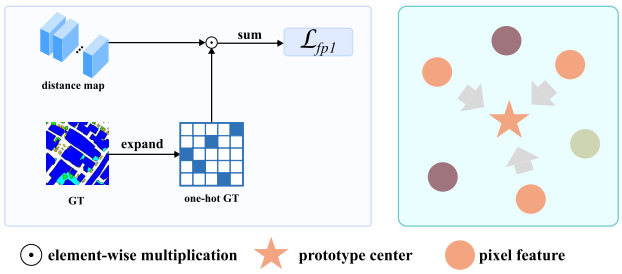


Fig. 4. Distance computation between pixels in $\mathcal{R}$ and the prototypes of ground-truth classes. Specifically, we apply a mask to the distance map (i.e., $\|\mathcal{F} - \mathcal{P}\|_2$) for obtaining the Euclidean distance between pixels to the corresponding prototype of class k, and we constrain the minimum Euclidean distance.

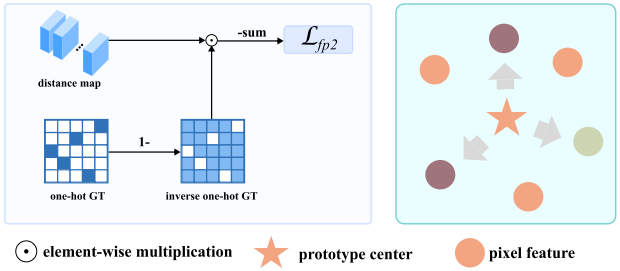


Fig. 5. Distance computation between pixels in $\mathcal{R}$ and the prototypes of nonground-truth classes. Specifically, we constrain the minimum distance from a pixel to a prototype of another class and hope it to be as large as possible.

a) Reduce the distance between pixel feature and the corresponding prototype: An intuitive idea is that we directly force the pixel to be equal to the most similar subprototype of the corresponding class. Specifically, as shown in Fig. 4, we apply a mask to $\|\mathcal{F} - \mathcal{P}\|_2$ to obtain the Euclidean distance between pixels and the corresponding prototype of class k. We choose the minimum distance to constrain

$$\mathcal{L}_{fp1} = \sum_{k=1}^{K} \frac{1}{\mathcal{N}(k)} \sum_{i=1}^{\mathcal{N}(k)} \Gamma^k \min\left(\|\mathcal{F}_i - \mathcal{P}_k\|_2\right) \tag{17}$$

where $\mathcal{N}(k)$ denotes pixel numbers of class k. Γ^k denotes the mask of class k, i.e., when pixel i belongs to class k, the value is 1, else 0. Therefore, intraclass compactness is achieved by approximating a pixel and the closest subprototype of its corresponding class to be equal.

b) Increase the distance between pixel feature and prototype of other class: Similarly, as shown in Fig. 5, we constrain the minimum distance from a pixel to a prototype of another class and hope it to be as large as possible. We also add an additional parameter ϵ to impose supervision

$$\mathcal{L}_{fp2} = \sum_{k=1}^{K} \frac{1}{\mathcal{N}(k)} \sum_{i=1}^{\mathcal{N}(k)} \Lambda^k \left(\epsilon - \min\left(\|\mathcal{F}_i - \mathcal{P}_k\|_2\right)\right) \tag{18}$$

where Λ^k denotes the reverse mask of class k, i.e., when pixel i belongs to class k, the value is 0, else 1. Thus, interclass separability is achieved by distancing pixels from the nearest subprototypes of other classes.

The final feats-to-prototype loss function ($\mathcal{L}_{fp}$) can be expressed as

$$\mathcal{L}_{fp} = \mathcal{L}_{fp_1} + \mathcal{L}_{fp_2}. \tag{19}$$

$\mathcal{L}_{fp}$ facilitates the joint embedding space of constrained pixels and prototypes. It also achieves intraclass compactness and interclass separability at the pixel level.

IV. Experiments

We conduct experiments on three widely used benchmark datasets for semantic segmentation of RSIs, which are detailed as follows.

A. Dataset

1) *ISPRS Vaihingen dataset* [43] is a small town in southern Germany. The main building types in Vaihingen are adjacent buildings and small multistory buildings with less traffic in the town. The ground sampling distance of the Vaihingen dataset is 9 cm. It contains 33 image blocks, of which 16 blocks are used for training, and the remaining 17 blocks are used for testing. The size of each image block is different, about 2000 × 2500 pixels. Six classes of objects, namely, building, car, low vegetation, impervious surface, tree, and background, are labeled.
2) *ISPRS Potsdam dataset* [43] is a typical historic city in northeastern Germany. It has large buildings, narrow streets, and dense traffic. The ground sampling distance of the Potsdam dataset is 5 cm. It contains 38 image blocks; among them, 23 blocks are used for training, and the remaining 14 blocks are used for testing. The size of each image block is 6000 × 6000 pixels. The required labeling classes are the same as the Vaihingen dataset.
3) *LoveDA* dataset [44] intrigued a new challenge in semantic segmentation of large-scale satellite images with a spatial resolution of 0.3 m. Acquired from the Google Earth Platform, LoveDA covers a total area of over 536 km^2. It includes both rural and urban areas of three cities, i.e., Nanjing, Changzhou, and Wuhan. Each image in the dataset has a spatial size of 1024 × 1024 pixels. In this study, we adopt 2522 images for training, 1669 images for validation, and 1796 images for testing. The class distribution is imbalanced, and congeneric objects vary in scale, size, and surface type, making LoveDA an even more challenging dataset.
4) *iSAID* dataset [45] consists of 2806 RSIs sourced from multiple satellites and sensors, with original image sizes ranging from 800 × 800 pixels to 4000 × 13 000 pixels. In addition, iSAID stands as one of the largest geospatial semantic segmentation datasets for remote sensing imagery, containing 655 451 densely annotated object instances across 15 categories within 2806 high-resolution images. The dataset is divided into predefined training, validation, and test sets, with 1411, 458, and 937 images, respectively.

B. Evaluation Metrics

In this study, we use commonly adopted evaluation metrics to measure the performance of the predicted results on the

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TABLE I
COMPARATIVE RESULTS ON THE ISPRS VAIHINGEN DATASET. PER-CLASS BEST PERFORMANCE IS MARKED IN BOLD

Method	Imp.	Bui.	Low.	Tree	Car	mIoU	F_1	OA
FarSeg++(TPAMI'23)	86.47	90.86	74.39	82.87	76.88	82.29	89.77	90.91
PEM(CVPR'24)	85.88	91.98	73.65	82.55	76.98	82.21	90.21	91.22
LOGCAN++(TGRS'25)	88.33	91.65	77.94	81.68	79.85	83.89	90.87	91.85
FarSeg(CVPR'20)	84.10	89.27	71.10	80.10	71.1	79.14	87.88	89.57
+ CenterSeg	**86.68**	**91.76**	**74.12**	**83.33**	**80.79**	**83.34**	**90.80**	**91.20**
ISNet(ICCV'21)	85.91	90.89	72.96	81.93	80.01	82.36	90.19	90.52
+ CenterSeg	**86.53**	**91.72**	**74.19**	**83.21**	**82.74**	**83.68**	**91.01**	**91.22**
ConvNeXt(CVPR'22)	87.13	91.56	73.81	83.48	78.37	82.87	90.50	91.36
+ CenterSeg	**88.39**	**92.24**	**75.18**	**84.32**	**79.73**	**83.97**	**91.32**	**91.68**
UNetFormer(ISPRS J'22)	88.14	91.47	71.23	81.46	79.22	82.30	90.31	90.82
+ CenterSeg	**89.28**	**93.42**	**72.19**	**82.80**	**80.44**	**83.63**	**91.21**	**91.42**
EfficientViT(ICCV'23)	87.26	90.42	69.14	78.41	77.37	80.52	87.56	89.41
+ CenterSeg	**89.96**	**93.44**	**72.37**	**80.65**	**80.13**	**83.31**	**91.52**	**91.23**
DOCNet(GRSL'24)	87.68	92.31	72.44	81.98	85.64	84.01	91.19	91.53
+ CenterSeg	**88.45**	**92.51**	**72.85**	**83.87**	**86.33**	**84.80**	**91.63**	**91.97**
SCSM(ISPRS J'25)	89.53	92.01	73.44	82.23	86.19	84.68	91.59	92.22
+ CenterSeg	**90.12**	**92.67**	**74.43**	**83.11**	**86.50**	**85.37**	**91.89**	**92.56**

TABLE II
COMPARATIVE RESULTS ON THE ISPRS POTSDAM DATASET. PER-CLASS BEST PERFORMANCE IS MARKED IN BOLD

Method	Imp.	Bui.	Low.	Tree	Car	mIoU	F_1	OA
FarSeg++(TPAMI'23)	87.77	93.88	77.21	79.64	92.65	86.23	91.99	91.22
PEM(CVPR'24)	86.86	93.68	77.34	79.65	90.23	85.55	91.73	90.48
LOGCAN++(TGRS'25)	88.16	94.65	81.55	80.68	92.81	87.57	93.11	91.48
FarSeg(CVPR'20)	85.28	91.33	76.46	77.36	91.37	84.36	91.21	89.87
+ CenterSeg	**88.14**	**94.39**	**78.63**	**80.76**	**93.43**	**87.07**	**92.96**	**91.54**
ISNet(ICCV'21)	87.53	93.68	78.60	80.03	93.06	86.58	92.67	91.27
+ CenterSeg	**88.54**	**94.92**	**78.70**	**80.90**	**93.90**	**87.39**	**93.14**	**91.82**
ConvNeXt(CVPR'22)	88.31	93.31	80.76	82.45	91.02	87.17	93.03	91.66
+ CenterSeg	**88.89**	**94.77**	**81.86**	**83.66**	**92.53**	**88.34**	**93.84**	**91.90**
UNetFormer(ISPRS J'22)	87.58	94.23	79.45	80.18	91.27	86.54	92.77	91.46
+ CenterSeg	**88.13**	**94.76**	**80.23**	**80.88**	**91.85**	**87.17**	**93.58**	**92.11**
EfficientViT(ICCV'23)	84.37	91.49	79.66	77.43	88.20	84.23	90.11	89.61
+ CenterSeg	**89.23**	**95.67**	**81.46**	**80.89**	**92.33**	**87.92**	**93.88**	**92.58**
DOCNet(GRSL'24)	88.43	94.97	81.57	81.86	91.47	87.64	93.50	92.02
+ CenterSeg	**88.98**	**95.56**	**82.01**	**82.66**	**92.50**	**88.34**	**94.03**	**92.43**
SCSM(ISPRS J'25)	88.44	94.48	82.55	79.76	93.72	87.79	93.60	92.13
+ CenterSeg	**89.11**	**95.47**	**83.40**	**80.23**	**93.99**	**88.44**	**94.23**	**92.54**

test set, including IOU class score, mean intersection over union (mIoU), overall accuracy (OA), and F_1 score. mIoU is calculated as a global metric to evaluate accuracy. OA measures the number of correctly classified pixels against all pixels. F_1 score is the harmonic mean of precision and recall, which evaluates the tradeoff between false positives and false negatives.

C. Implementation Details

Our model is implemented based on the PyTorch library. All experiments are conducted using one NVIDIA A6000 GPU. The training patches are randomly sampled with the patch size of 512 $\times$ 512 and the batch size of 4. The stochastic gradient descent (SGD) optimizer is used with the learning rate initialized to 0.01 with a weight decay of 0.0001. We choose the 0.999 as the momentum coefficient. The model is trained for 150 epochs on the Vaihingen dataset, 100 epochs on the Potsdam dataset, and 50 epochs on the LoveDA dataset.

D. Comparison With State-of-the-Art Methods

The proposed CenterSeg is a plug-and-play classifier that could facilitate the arbitrary segmentation network to optimize. The CenterSeg is employed as the commonly used high-resolution RSI semantic segmentation networks, including FarSeg [15], ISNet [46], UNetFormer [14], DOCNet [47], and SCSM [31]. In addition, we also use the recent proposed backbone, e.g., ConvNeXt [48], SwinTransformer [41], [49], and EfficientViT [50]. To further illustrate the effects of

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TABLE III

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TEST SET OF THE LOVEDA DATASET. PLEASE NOTE THAT THE LOVEDA DATASET REQUIRES AN ONLINE TEST TO EVALUATE THE MODEL. THEREFORE, RESULTS FOR THE F_1 AND OA METRICS ARE NOT AVAILABLE HERE. PER-CLASS BEST PERFORMANCE IS MARKED IN BOLD

Method	Back.	Bui.	Road	Bar.	Was.	For.	Agr.	mIoU
FarSeg++ (TPAMI'23)	44.1	57.7	54.2	78.3	11.1	40.6	59.9	49.4
PEM (CVPR'24)	43.3	56.7	53.2	79.4	14.3	41.7	58.3	49.6
LOGCAN++ (TGRS'25)	47.4	58.4	56.5	80.1	18.4	47.9	64.8	53.4
FarSeg (CVPR'20)	43.1	51.5	53.9	76.6	9.8	43.3	58.9	48.2
+ CenterSeg	**44.5**	**57.0**	53.1	**78.8**	12.5	**47.3**	**63.9**	**51.0**
ISNet (ICCV'21)	44.4	57.4	58.0	77.5	**21.8**	43.9	60.6	51.9
+ CenterSeg	**46.3**	**58.6**	**60.0**	**78.8**	**23.4**	**44.5**	**62.3**	**53.4**
ConvNeXt (CVPR'22)	46.9	53.5	56.8	76.1	15.9	47.5	61.8	51.2
+ CenterSeg	**47.2**	55.0	55.1	**79.4**	**16.8**	**48.6**	**64.3**	**52.3**
UNetFormer (ISPRS J'22)	44.7	58.8	54.9	79.6	20.1	46.0	62.5	52.4
+ CenterSeg	**45.8**	**59.7**	**56.2**	**79.9**	**21.5**	**47.8**	**63.5**	**53.5**
EfficientViT (ICCV'23)	42.9	51.0	52.8	75.7	4.3	42.0	61.2	47.1
+ CenterSeg	**43.7**	**54.3**	**55.2**	**78.1**	**10.8**	**47.4**	**63.0**	**50.4**
DOCNet (GRSL'24)	46.6	59.3	56.9	81.5	21.3	46.2	65.2	53.9
+ CenterSeg	47.6	**60.7**	**57.9**	**81.6**	**22.5**	**47.8**	66.1	**54.9**
SCSM (ISPRS J'25)	48.3	60.4	58.4	80.7	19.6	47.6	67.2	54.6
+ CenterSeg	**49.4**	**61.2**	**59.4**	**80.9**	**21.0**	**47.9**	**67.9**	**55.4**

TABLE IV

COMPARATIVE RESULTS ON THE iSAID DATASETS. PER-CLASS BEST PERFORMANCE IS MARKED IN BOLD, AND THE SECOND LARGEST VALUE IS UNDERLINED

Method	IoU															mIoU
	Ship	ST	BD	TC	BC	GTF	Br.	LV	SV	HC	SP	RA	SBF	Pl.	Ha.	
NonLocal [13]	63.4	48.0	49.5	86.4	62.7	50.0	35.0	57.7	43.4	31.6	44.9	67.4	71.0	80.0	51.5	58.8
DANet [37]	63.9	46.2	73.7	85.7	57.9	48.2	33.5	57.9	43.2	36.1	45.7	67.2	69.2	80.4	52.3	60.0
PSPNet [9]	65.2	52.1	75.7	85.5	61.1	60.1	32.4	58.0	42.9	10.8	46.7	68.6	71.9	79.5	54.2	60.2
EMANet [38]	65.3	52.8	72.2	86.0	62.8	49.0	34.9	57.6	43.1	38.6	46.0	69.2	69.3	80.7	52.8	61.2
DeepLabv3+ [39]	63.7	58.4	75.6	86.5	59.9	58.6	34.9	59.1	43.9	27.9	48.2	68.7	74.5	80.3	51.9	61.9
HRNet [40]	67.2	64.4	78.2	87.6	60.9	57.5	34.8	59.9	47.7	15.9	48.9	68.2	74.5	82.3	57.0	62.7
UperNet [41]	65.9	59.0	75.7	87.1	61.6	58.5	36.1	60.0	45.7	33.6	49.5	70.6	73.5	81.7	54.4	63.2
FarSeg [15]	65.3	61.8	77.7	86.3	62.0	56.7	36.7	60.5	46.3	35.8	51.2	71.3	72.5	82.0	53.9	63.7
LOGCAN++ [19]	68.4	70.2	78.8	87.6	66.3	61.1	44.3	63.2	50.8	38.9	52.9	74.3	74.7	84.3	59.1	65.0
+CenterSeg	69.4	70.9	79.5	88.1	66.6	61.8	46.1	63.8	51.5	39.6	53.4	74.4	75.6	84.8	59.8	65.7

CenterSeg, some recent state-of-the-art models, such as FarSeg++ [16], PEM [51], and LOGCAN++ [19], are used only for comparison.

1) Quantitative Analysis: The results are listed in Tables I–IV, which shows that the CenterSeg significantly improves the performance of different RSI segmentation models in three datasets. With the introduction of the CenterSeg, a high accuracy improvement can be obtained for all the models. In the Vaihingen dataset, CenterSeg is more conducive to improving the segmentation performance of small objects (e.g., trees and cars). In addition, adding CenterSeg performs relatively better on the "low vegetation" in the Potsdam dataset, which lies in the fact that "low vegetation" has more characteristic representations, i.e., large intraclass variance. The LoveDA is a challenging dataset because of the more obvious intraclass diversity and interclass similarity. Table III shows that the CenterSeg also has superior performance on the LoveDA dataset. Overall, CenterSeg achieves state-of-the-art segmentation performance on all three datasets compared to recent popular methods, such as PEM [51], FarSeg++ [16], and LOGCAN++ [19]. Although the absolute improvement is between 0.5% and 1.8%, on a complex dataset like iSAID containing 15 categories (shown in Table IV), it achieves a comprehensive improvement over the top-tier method LOGCAN++, which is already nearing performance saturation, demonstrating the effectiveness of CenterSeg as a classifier module. It is particularly noteworthy that CenterSeg shows a more significant improvement on categories with high intraclass variance [such as bridges (BD) and helicopters (HC)], which is consistent with our design intent.

2) Visualization Analysis: Figs. 6–8 provide additional visual comparisons on the employed three datasets, where DeepLabv3 +, DANet, FarSeg, ISNet, ConvNeXt, HRNet, and UperNer are selected as the baseline models. We use dashed boxes for emphasis on the focus regions. The results show that CenterSeg can obtain better visualization performance. Specifically, CenterSeg is able to fragment small objects (e.g., car classes) completely. This proves the effectiveness of CenterSeg. For some objects, such as "car" in the Vaihingen dataset and "tree" in the Potsdam dataset, SwinUperNet can distinguish these categories. However, several objects, such as "road" and "forest" in the LoveDA dataset, are vaguely assigned because the classes are geographically close to each other. In addition, "building" in the Potsdam

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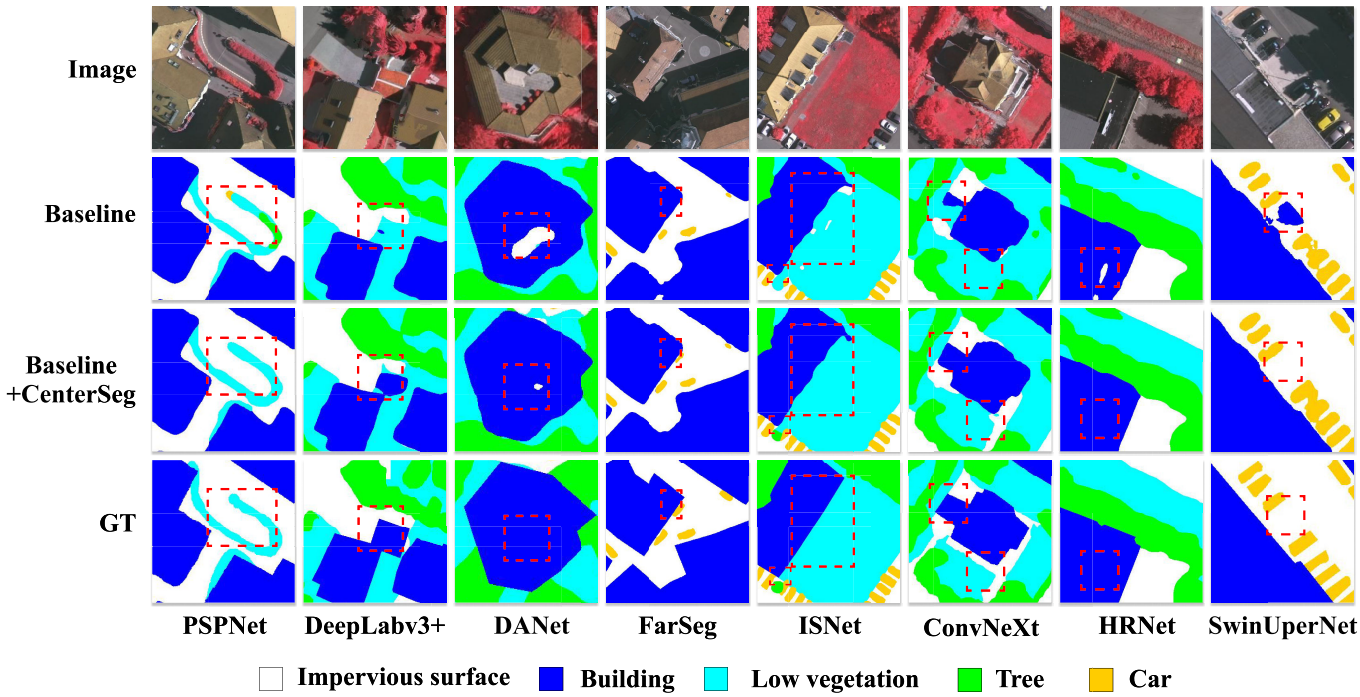


Fig. 6. Visual comparison with several baselines on the ISPRS Vaihingen test set. The red dashed box is the area of focus. Best viewed in color and zoomed-in view.

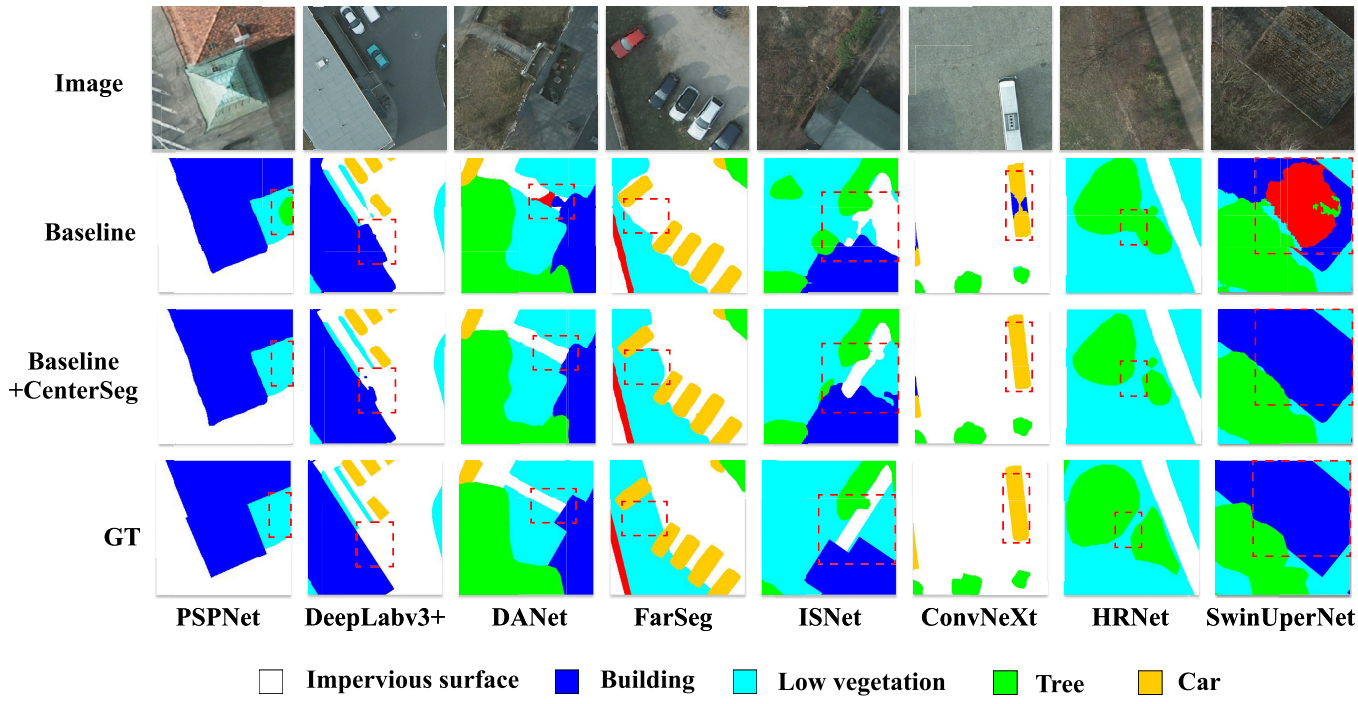


Fig. 7. Visual comparison with several baselines on the Potsdam test set. The red dashed box is the area of focus. Best viewed in color and zoomed-in view.

dataset is incorrectly classified as "clutter" due to the extremely similar semantic features. In contrast, the SwinUperNet (CenterSeg) can obtain more local and global feature representations by adding the proposed prototype classifier. Thus, geo-objects are correctly identified whether they are natural objects (e.g., tree) or artificial objects (e.g., building and road).

3) Comparison With Existing Classifiers: As shown in Table V, we compare with existing classifiers (designed for natural images) on the LoveDA dataset. It can be observed

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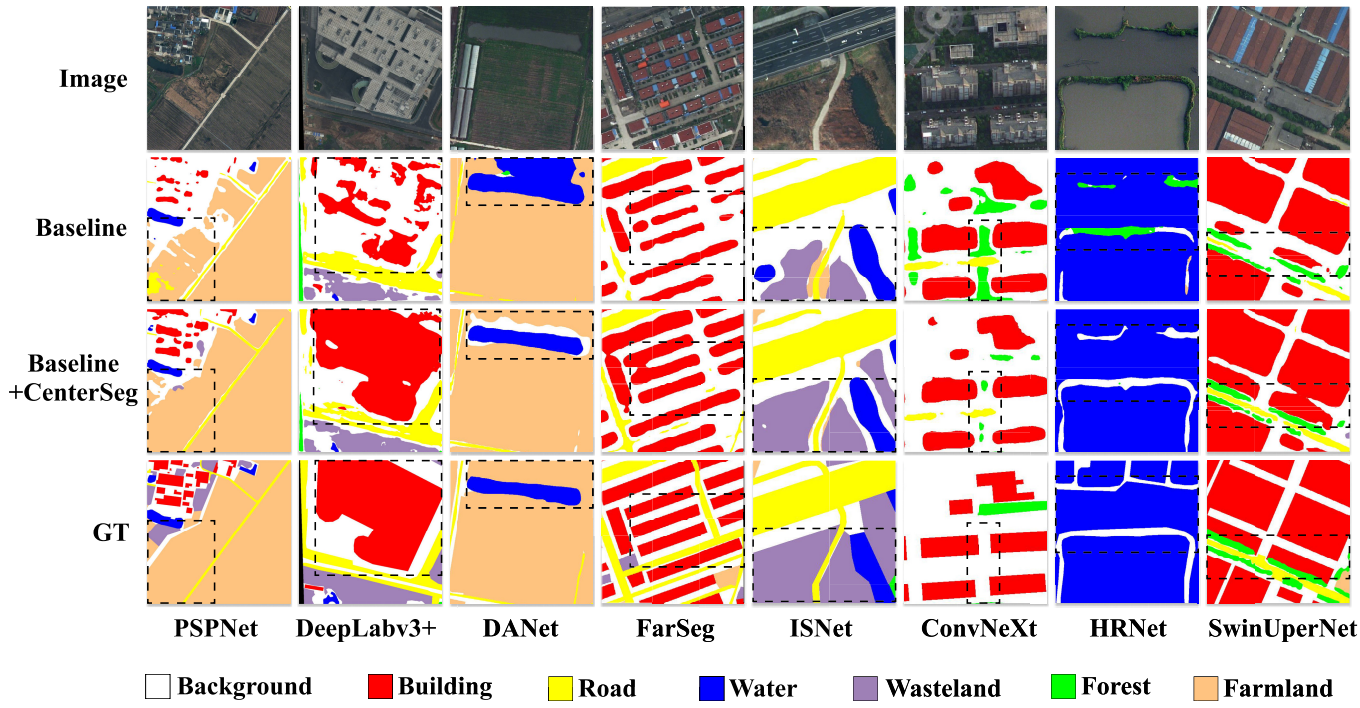


Fig. 8. Visual comparison with several baselines on the LoveDA test set. The red dashed box is the area of focus. Best viewed in color and zoomed-in view.

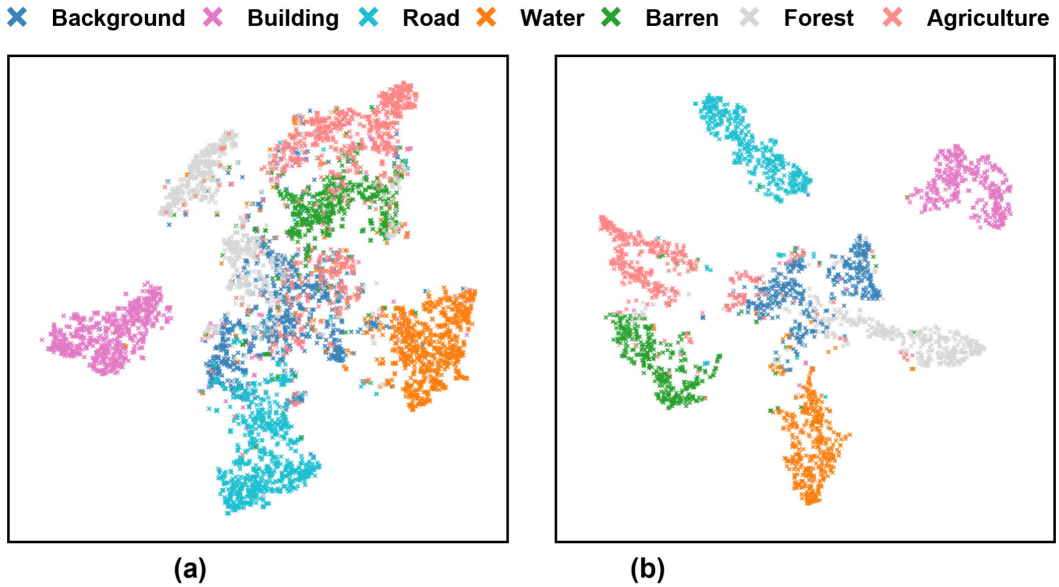


Fig. 9. Visualization of class features distributions by the last layer of (a) ConvNeXt and (b) ConvNeX+CenterSeg. The test image is selected from the LoveDA dataset. We implement the experiment with t-SNE [42].

that CenterSeg achieves better segmentation performance and efficiency, which is attributed to our customized class-centered generative multiprototype strategy and more appropriate regularization terms for RSIs.

4) Class Feature Distributions: Fig. 9 illustrates the class feature distributions of ConvNeXt and ConvNeXt+CenterSeg by using t-SNE [42]. It shows that the distinction between each class is not obvious in ConvNeXt, as shown in Fig. 9(a). In contrast, ConvNeXt+CenterSeg shows a more compact distribution in the feature space than ConvNeXt in certain classes, e.g., forest and road. Furthermore, as shown in Fig. 10, the features generated by ConvNeXt+CenterSeg (final layer) exhibit more compact clusters within classes, while pixel features from different classes remain distinctly separated.

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TABLE V

COMPARISON WITH EXISTING CLASSIFIERS WITH EVALUATION METRICS, INCLUDING PARAMETERS (M), FLOPS (G), FPS, MEMORY (GB), TRAINING TIME (HOURS), AND MIOU

Method	Params	FLOPs	fps	Memory	Training Time	mIoU
ConvNeXt(CVPR'22)	59.3	235	15.7	5.5	4.5	51.2
+ ProtoSeg(CVPR'22)	59.2	237	15.2	8.5	12.0	51.8
+ GMMSeg(NeurIPS'22)	59.2	236	15.4	7.8	13.3	51.7
+ SSA-Seg(NeurIPS'24)	59.3	235	15.6	5.6	8.8	51.8
+ CenterSeg	59.2	236	15.4	5.6	5.3	52.3

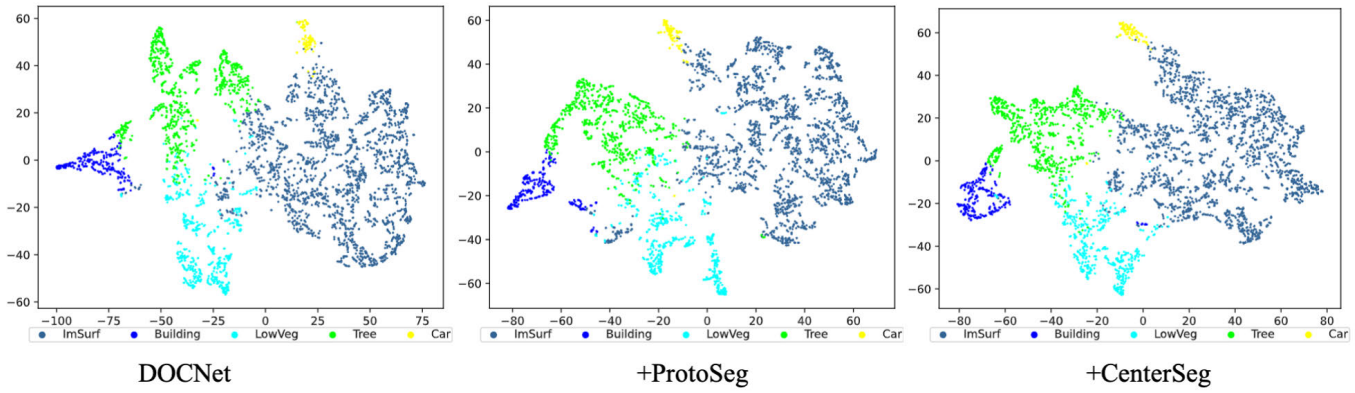


Fig. 10. Visualization of class features distributions by the DOCNet (baseline), DOCNet+ProtoSeg, and DOCNet+CenterSeg. The test image is selected from the Vaihingen dataset. We implement the experiment with t-SNE [42].

V. ABLATION STUDY

We perform ablation studies on the ISPRS Vaihingen dataset. The ConvNeXt is selected as the baseline in all studies.

A. Class Activation Map

As shown in Fig. 11, we set car, building, and tree as the target classes to obtain the corresponding class activation map based on Grad-CAM [52]. It can be observed that the CenterSeg has higher activations in the target classes, as well as concentrates more on the building/tree/car area than the Baseline (ConvNeXt).

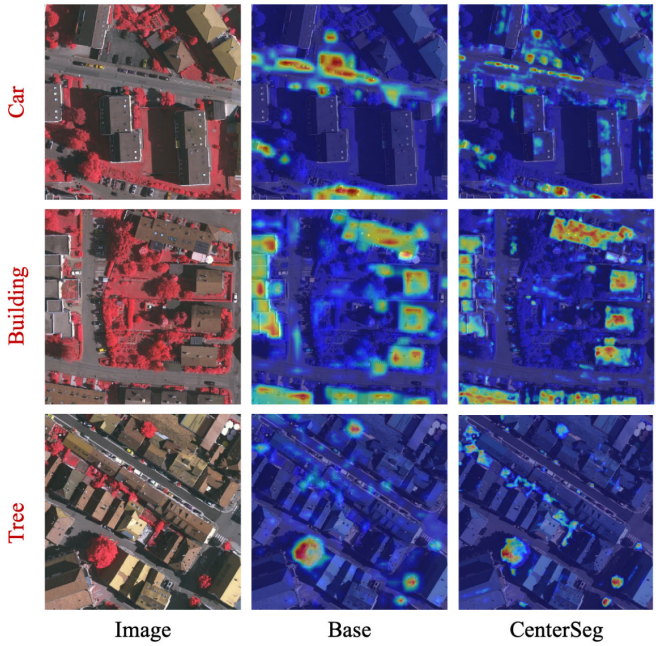


Fig. 11. Class activation maps for CenterSeg and baseline (ConvNeXt). Images are selected from the ISPRS Vaihingen dataset. From top to bottom, the target classes are car, building, and tree, respectively.

B. Prototype Interpretability

Prototype-based classifiers are favored for their interpretability over parametric softmax classifiers. Then, there is no related research work in the field of semantic segmentation of RSIs. Our approach is, to some extent, inspired by ProtoPNet [26], ProtoSeg [24], and GMMSeg [23]. Specifically, by approximating the prototype to an image patch on the training set, we can perform classification decisions based on *this looks like that* rule. This classification rule provides an intuitive interpretation mechanism for the model, helping us to understand the basis of the model's decisions. Furthermore, by observing and analyzing the prototype, we can infer the model's focus on different categories and distinguishing features, and thus explain the model's predictions.

As shown in Figs. 12–14, we perform interpretable validation on three datasets. As can be seen, CenterSeg captures well-characterized prototypes and can perform classification based on these prototypes. This gives CenterSeg clear decision rules and provides strong interpretability. To the best of our knowledge, there has been no work on interpretability in the field of RSI segmentation.

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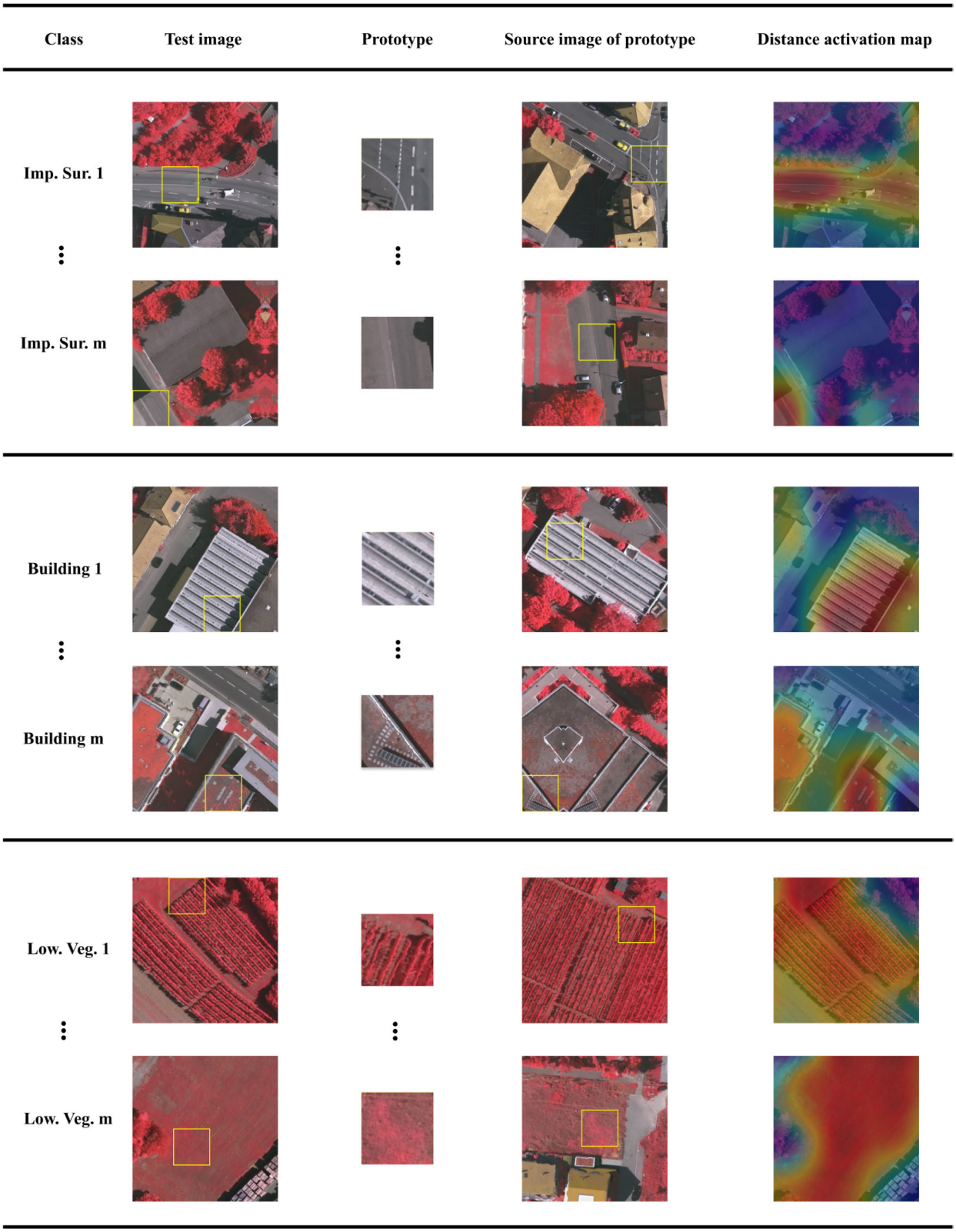


Fig. 12. Prototype interpretability analysis for impervious surface class, building class, and low vegetation class. Images are from the ISPRS Vaihingen dataset.

C. Number of Prototypes for Per-Class

As shown in Table VI, mIoU rises as the prototype number increases, and reaches the maximum (i.e., 83.97) when the prototype number is 8. However, the segmentation results decrease slightly when the prototype number is 16. The reason is significant prototype number is fixed in a certain dataset. The mIoU will no longer change significantly when the set prototype number is greater than the significant prototype number.

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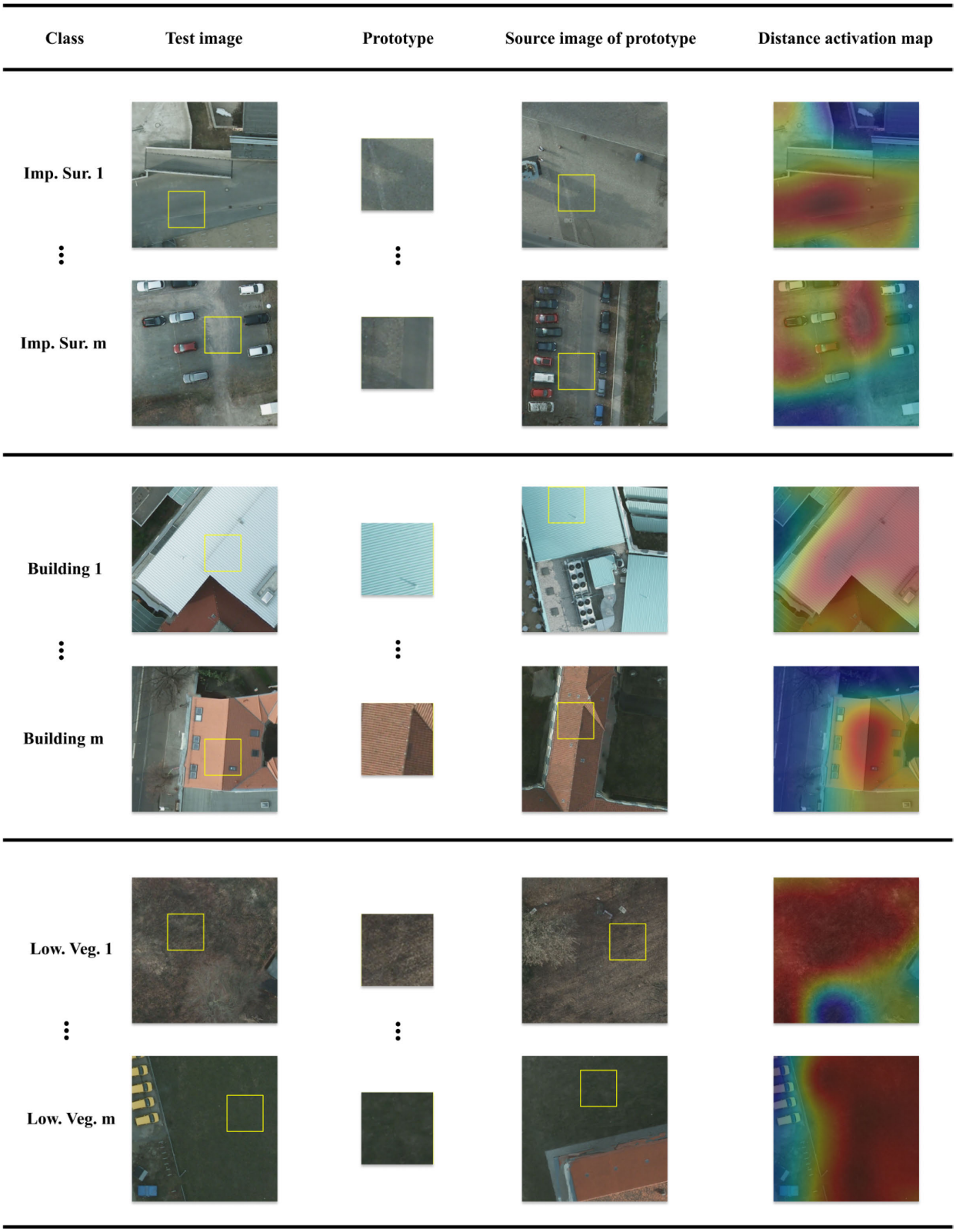


Fig. 13. Prototype interpretability analysis for impervious surface class, building class, and low vegetation class. Images are from the ISPRS Potsdam dataset.

D. Effectiveness of Grassmann Manifold Regularization and HAA

Compared to the proposed model (CenterSeg), the performance of CenterSeg without Grassmann manifold regularization (GMR) is significantly degraded, as shown in Table VIII. This clearly demonstrates that the GMR loss term itself is crucial for learning more discriminative center point features. Removing it weakens the feature space constraints, reduces the ability to represent center points, and thus affects segmentation accuracy. Similarly, compared to the CenterSeg,

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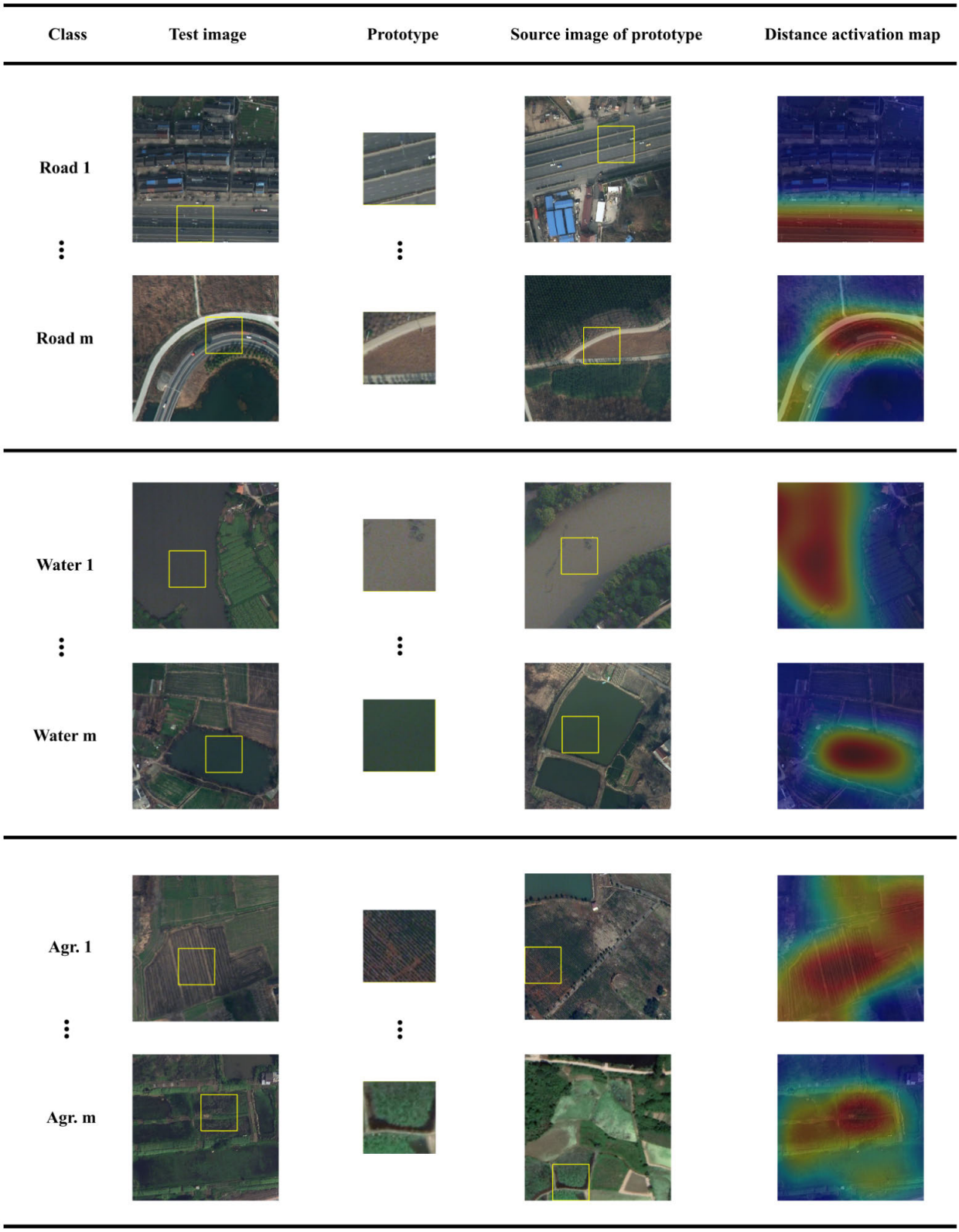


Fig. 14. Prototype interpretability analysis for road class, water class, and agriculture class. Images are from the LoveDA dataset.

the performance of CenterSeg without HAA only shows a significant drop, as shown in Table IX. This strongly proves that the HAA is not redundant, but a key component to improve model performance.

E. Combination of Loss Function

As can be seen from Table VII, the proposed regularization terms $\mathcal{L}_{pp}$ and $\mathcal{L}_{fp}$ have positive feedback. Specifically, $\mathcal{L}_{pp}$

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TABLE VI
EFFECT OF THE PATCH SIZE AND PROTOTYPE NUMBERS PER-CLASS

patch size	m	mIoU
1×1	1	81.60
2×2	1	78.64
2×2	4	83.16
2×2	8	83.97
2×2	16	83.35
4×4	8	82.12
8×8	8	68.35

TABLE VII
EFFECT OF THE COMBINATION OF LOSS FUNCTION

$\mathcal{L}_{ce}$	$\mathcal{L}_{Dice}$	$\mathcal{L}_{pp}$	$\mathcal{L}_{fp}$	mIoU
✓	✓			82.90
✓	✓	✓		83.42
✓	✓		✓	83.36
✓		✓	✓	83.76
✓	✓	✓	✓	83.97

TABLE VIII
ABLATION STUDY OF GMR

Model	mIoU	OA	F_1
CenterSeg	83.97	91.68	91.32
CenterSeg w/o GMR	83.21	91.57	91.04

TABLE IX
ABLATION STUDY OF HAA

Model	mIoU	OA	F_1
CenterSeg	83.97	91.68	91.32
CenterSeg w/o HAA	83.76	91.52	91.18

TABLE X
ABLATION STUDY OF $\mathcal{L}_{pp}$

Baseline	$\mathcal{L}_{pp_1}$	$\mathcal{L}_{pp_2}$	mIoU
✓			82.90
✓	✓		83.57
✓		✓	83.46
✓	✓	✓	83.97

has the greatest impact on the model as the orthogonality between class prototypes ensures the diversity of class prototypes (see line 2). In addition, $\mathcal{L}_{fp}$ has a significant improvement in the baseline model because of increasing the intraclass compactness of the baseline (see line 3). The model achieves the best mIoU (83.97%) when using both proposed regularizer terms.

In addition, we separately test the two important components of the regularization term $\mathcal{L}_{pp}$, namely $\mathcal{L}_{pp_1}$ and $\mathcal{L}_{pp_2}$, as shown in Table X. It can be seen that the two components have a positive effect on the segmentation effect. Table XI shows the ablation experiment of the two important components of the regularization term $\mathcal{L}_{fp}$, which also demonstrates the effectiveness of these two components.

TABLE XI
ABLATION STUDY OF $\mathcal{L}_{fp}$

Baseline	$\mathcal{L}_{fp_1}$	$\mathcal{L}_{fp_2}$	mIoU
✓			82.90
✓	✓		83.27
✓		✓	83.76
✓	✓	✓	83.97

VI. CONCLUSION

Considering the characteristic of RSIs, i.e., large intraclass variance, we propose a novel and efficient classifier (called CenterSeg) for RSI semantic segmentation. Specifically, CenterSeg extracts local class centers for generating prototypes. In addition, to optimize the generation of prototypes, we introduce the Grassmann manifold and constrain the joint embedding space of pixel features and prototypes based on two additional regular terms. Experiments demonstrate that CenterSeg has the advantages of effectiveness, simplicity, lightweight, compatibility, and interpretability, which reveal its potential for RSI segmentation tasks. We hope that this work will promote more classifier-related research for RSI segmentation tasks.

REFERENCES

[1] J. Geng, S. Song, and W. Jiang, "Dual-path feature aware network for remote sensing image semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 5, pp. 3674–3686, May 2024.
[2] A. Samie et al., "Examining the impacts of future land use/land cover changes on climate in Punjab province, pakistan: Implications for environmental sustainability and economic growth," *Environ. Sci. Pollut. Res.*, vol. 27, no. 20, pp. 25415–25433, Jul. 2020.
[3] Q. Zhang and K. C. Seto, "Mapping urbanization dynamics at regional and global scales using multi-temporal DMSP/OLS nighttime light data," *Remote Sens. Environ.*, vol. 115, no. 9, pp. 2320–2329, 2011.
[4] A. Milioto, P. Lottes, and C. Stachniss, "Real-time semantic segmentation of crop and weed for precision agriculture robots leveraging background knowledge in CNNs," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2018, pp. 2229–2235.
[5] C. Yang, J. Wang, H. Meng, S. Yang, and Z. Feng, "Negative class guided spatial consistency network for sparsely supervised semantic segmentation of remote sensing images," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 1, pp. 657–669, Jan. 2025.
[6] Q. Zang, S. Wang, D. Zhao, Z. Zhong, B. Hou, and L. Jiao, "Joint style and layout synthesizing: Toward generalizable remote sensing semantic segmentation," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 5, pp. 4055–4071, May 2025.
[7] L. Dai, G. Zhang, and R. Zhang, "RADANet: Road augmented deformable attention network for road extraction from complex high-resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–13, 2023.
[8] H. Jung, H.-S. Choi, and M. Kang, "Boundary enhancement semantic segmentation for building extraction from remote sensed image," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–12, 2022.
[9] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Honolulu, HI, USA, Jul. 2017, pp. 6230–6239.
[10] Y. Yuan, X. Chen, and J. Wang, "Object-contextual representations for semantic segmentation," in *Proc. ECCV*, Aug. 2020, pp. 173–190.
[11] R. Strudel, R. Garcia, I. Laptev, and C. Schmid, "Segmenter: Transformer for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7242–7252.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 06,2026 at 22:19:42 UTC from IEEE Xplore. Restrictions apply.

[12] E. Xie, W. Wang, Z. Yu, A. Anandkumar, J. M. Alvarez, and P. Luo, "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 12077–12090.

[13] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7794–7803.

[14] L. Wang et al., "UNetFormer: A UNet-like transformer for efficient semantic segmentation of remote sensing urban scene imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 190, pp. 196–214, Aug. 2022.

[15] Z. Zheng, Y. Zhong, J. Wang, and A. Ma, "Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, 2020, pp. 4095–4104.

[16] Z. Zheng, Y. Zhong, J. Wang, A. Ma, and L. Zhang, "FarSeg++: Foreground-aware relation network for geospatial object segmentation in high spatial resolution remote sensing imagery," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 11, pp. 13715–13729, Nov. 2023.

[17] X. Ma, Z. Ni, and X. Chen, "TinyViM: Frequency decoupling for tiny hybrid vision Mamba," 2024, *arXiv:2411.17473*.

[18] R. Zuo, G. Zhang, R. Zhang, and X. Jia, "A deformable attention network for high-resolution remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022.

[19] X. Ma et al., "LOGCAN++: Adaptive local–global class-aware network for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025.

[20] X. Ma et al., "Log-can: Local–global class-aware network for semantic segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Jun. 2023, pp. 1–5.

[21] X. Ma et al., "SACANet: Scene-aware class attention network for semantic segmentation of remote sensing images," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2023, pp. 828–833.

[22] W. Wang, H. Cheng, T. Zhou, and D. Liu, "Visual recognition with deep nearest centroids," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Kigali, Rwanda, May 2023.

[23] L. Chen, W. Wang, J. Miao, and Y. Yang, "GMMSeg: Gaussian mixture based generative semantic segmentation models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2022, pp. 31360–31375.

[24] T. Zhou, W. Wang, E. Konukoglu, and L. Van Goo, "Rethinking semantic segmentation: A prototype view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 2582–2593.

[25] X. Chen, X. Ma, and Z. Ni, "SSA-seg: Semantic and spatial adaptive pixel-level classifier for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2024, pp. 93690–93713.

[26] C. Chen, O. Li, C. Tao, A. J. Barnett, J. K. Su, and C. Rudin, "This looks like that: Deep learning for interpretable image recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018.

[27] Y. Huang et al., "CARD: Semantic segmentation with efficient class-aware regularized decoder," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 10, pp. 9024–9038, Oct. 2024.

[28] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[29] L. Lv and L. Zhang, "Advancing data-efficient exploitation for semi-supervised remote sensing images semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024.

[30] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[31] X. Ma et al., "A novel scene coupling semantic mask network for remote sensing image segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 221, pp. 44–63, Mar. 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0924271625000255

[32] X. Li et al., "PointFlow: Flowing semantics through points for aerial image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 4215–4224.

[33] F. Yang and C. Ma, "Sparse and complete latent organization for geospatial semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1809–1818.

[34] E. Jang, S. Gu, and B. Poole, "Categorical reparameterization with gumbel-softmax," 2016, *arXiv:1611.01144*.

[35] A. Van Den Oord et al., "Neural discrete representation learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.

[36] U. Helmke, K. Hüper, and J. Trumpf, "Newton's method on Graßmann manifolds," 2007, *arXiv:0709.2205*.

[37] J. Fu et al., "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit.*, 2019, pp. 3146–3154.

[38] X. Li, Z. Zhong, J. Wu, Y. Yang, Z. Lin, and H. Liu, "Expectation-maximization attention networks for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2019, pp. 9166–9175.

[39] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder–decoder with atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 801–818.

[40] J. Wang et al., "Deep high-resolution representation learning for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 10, pp. 3349–3364, Oct. 2021.

[41] T. Xiao, Y. Liu, B. Zhou, Y. Jiang, and J. Sun, "Unified perceptual parsing for scene understanding," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 432–448.

[42] L. V. D. Maaten and G. E. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 86, pp. 2579–2605, Jan. 2008.

[43] F. Rottensteiner et al., "ISPRS semantic labeling contest," *ISPRS*, vol. 1, no. 4, p. 4, 2014.

[44] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," in *Proc. Neural Inf. Process. Syst.*, Aug. 2021.

[45] S. W. Zamir et al., "ISAID: A large-scale dataset for instance segmentation in aerial images," in *Proc. IEEE/CVF Conf. Comput. Vis. pattern Recognit. workshops*, 2019, pp. 28–37.

[46] Z. Jin, B. Liu, Q. Chu, and N. Yu, "ISNet: Integrate image-level and semantic-level context for semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 7169–7178.

[47] X. Ma et al., "DOCNet: Dual-domain optimized class-aware network for remote sensing image segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.

[48] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A ConvNet for the 2020s," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 11976–11986.

[49] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.

[50] H. Cai, J. Li, M. Hu, C. Gan, and S. Han, "EfficientViT: Lightweight multi-scale attention for high-resolution dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 17256–17267.

[51] N. Cavagnero et al., "PEM: Prototype-based efficient MaskFormer for image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2024, pp. 15804–15813.

[52] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Int. J. Comput. Vis.*, vol. 128, no. 2, pp. 336–359, Feb. 2020.

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